DANIEL LANDI

AI & Infrastructure Engineer | Local LLMs · Cloud · Systems Programming

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PROFESSIONAL SUMMARY

AI and infrastructure engineer with 25+ years in telecom operations — NOC engineering in Brazil and operational-process projects for carriers in six other countries — now building AI applications and local-LLM tooling. Recent work: a C inference runtime on llama.cpp for 8 GB, CPU-only hardware, with a Python eval harness that measures what small models can build; a self-correcting LLM activity generator containerized with Docker and deployed on Railway; and a failover-resilient serverless AWS architecture. AWS Certified Cloud Practitioner; completing the 42 São Paulo systems-programming curriculum. Brings carrier-grade operational discipline — SLAs, failure-first design, documentation someone else can act on at 3 AM — to AI systems that have to keep working.

CORE SKILLS

- **AI & LLM engineering:** local inference with llama.cpp (C API, GGUF, sampling and decoding); model evaluation with human-written judges (pass rates, retry strategies); LLM application design (prompt pipelines, self-critique loops); Groq-hosted Llama 3; AWS Bedrock and Textract; IBM Bob 2.0 CLI
- **Programming & tooling:** C (42 Norm), Python, Node.js/Express, Bash, Git/GitHub, Makefiles, Docker, manual memory management
- **Cloud & infrastructure:** AWS: Lambda, API Gateway, DynamoDB, S3, CloudFront, WAF, SQS, SNS, CloudWatch, IAM; Terraform (environment audit); Linux/Unix (Pop!_OS, Solaris, HP-UX)
- **Operations & process:** ITIL, eTOM, SLA and uptime management, NOC operations, SOPs and technical documentation

AI & SOFTWARE PROJECTS

uLLAMA — Local LLM Inference Runtime in C and Eval Harness | Personal project, Sep 2026 – Present

- Wrote a C runtime on the llama.cpp API that loads GGUF models and runs multi-turn chat with streamed output, configurable sampling (top-k, top-p, temperature) and live tokens-per-second measurement; designed to fit 8 GB RAM on CPU only, written under the 42 Norm.
- Built a Python eval harness: a small model gets a task, a human-written judge tests its code, and the harness retries and counts how often it passes. Measured that spelling out the approach in the prompt raised the single-attempt pass rate from about 3% to about 32% on one task. Reading passing code exposed a file-handle leak the judge missed, so the judge was tightened: passing a judge is only a lower bound.
- Benchmarked a harness-built binary search over a 100-million-record (2.4 GB) file: about 60 µs warm; a C/mmap version reached about 1.5 µs with the file already mapped. The repository documents what was and was not established. Code: github.com/drlandi/ullama

Self-Correcting AI Teaching Activity Generator | IBM Bob 2.0 Hackathon, Sep 2026

- Built and deployed, during a 48-hour hackathon, a Node.js/Express web app that generates hands-on, teacher-editable learning activities adapted to a classroom's resource level, with a second-pass self-critique loop that reviews and regenerates output against quality checks.
- Integrated the Bob CLI through child processes, resolved stream-handling hangs, added print-ready PDF and flowchart views, and containerized the app with Docker for a permanent Railway deployment. Validated against classroom activities I had already designed at CIEBP. Code: github.com/drlandi/ibm-bob-activity-generator

Digital Resilience Cloud Architecture | EDN Hackathon "Future Builders" (Escola da Nuvem × AWS), Sep 2026

- Designed, built and live-tested a failover-resilient serverless architecture: WAF and CloudFront with dual-origin failover (S3 in sa-east-1 and us-east-1), API Gateway with direct-to-SQS integration, Lambda worker, DynamoDB, CloudWatch/SNS.
- Validated end-to-end with real submissions and audited a teammate's Terraform-provisioned environment against requirements.

Insurance-Claims AI Prototypes | AWS Hackathon and personal project, 2026

- "Elementum": serverless claims-intake pipeline on S3, Lambda and DynamoDB, with Textract/Bedrock integration (mocked due to account verification constraints). "ClaimsCopilot": natural-language claim-analysis assistant using Groq-hosted Llama 3.

Systems Programming | 42 São Paulo — Software Engineering Cadet, 2025 – Present

- libft (43 C standard-library functions), get_next_line (static memory; fixed malloc-failure edge cases and two leaks), push_swap (co-developed adaptive sorting for large data sets), ft_printf (variadic arguments).

PROFESSIONAL EXPERIENCE

Specialist Teacher (Professor Especialista) | CIEBP — São Paulo State Public School — Feb 2023 – Present

- Design and teach activity series that bring computational thinking (unplugged programming) and AI literacy to classrooms with limited access to technology; build AI-assisted tooling to generate and quality-check classroom activities.

Electrical Engineering Consultant | TAUPO – Consultoria Ambiental — Nov 2015 – Dec 2021

- Delivered low-voltage building electrical projects and solar-energy projects in rural areas of northwestern Minas Gerais.

Maintenance Operations Coordinator | SEICOM (Vivo network, Espírito Santo) — Nov 2014 – Mar 2015

- Coordinated preventive and corrective maintenance of RF and infrastructure for Vivo's cellular sites; managed the technical and administrative team, reporting and contract targets.

Technical Education & Curriculum Leadership | UFES, Multivix, Master Grupo Educacional — 2010 – 2013

- Lecturer, Universidade Federal do Espírito Santo, São Mateus (Dec 2011 – Sep 2013): thermodynamics, strength of materials and solid mechanics for four engineering programs. Lecturer, Multivix (2011 – 2013): physics and solid mechanics.
- Technical Course Coordinator, Master Grupo Educacional (Nov 2010 – Jun 2013): Electrotechnics, Industrial Automation, Mechanics and Oil & Gas programs — operating procedures, teaching labs, course materials, hiring and managing faculty.

Telecom Specialist | Ericsson Telecomunicações Ltda. — Nov 2005 – Jul 2010

- International specialist on GSM and 3G (WCDMA) projects for carriers in six countries outside Brazil, designing and implementing operational processes under eTOM and ITIL.
- Zain (Nigeria): managed the Field Operations sub-project and new operating procedures under eTOM. Hutchison 3G (UK, Australia) and Etisalat (Egypt): contractual assessment and development of operational processes.
- Nawras (Oman): network configuration management project. EGS (Brazil): field-operations management system. Telkomsel and Natrindo (Indonesia): operational-process implementation.

Earlier career in network operations — 1991 – 2005

- NOC Engineer at Nextel (1998 – 2005) and Telesp Celular (1993 – 1998): maintenance of cellular switches (Nortel, NEC, Ericsson), signaling, messaging and prepaid systems, performance analysis, drive tests, operating procedures; Unix administration (Solaris, HP-UX) and design of a Windows NT 4.0 LAN.
- Telecommunications Technician at Bastec and Telepatch (1991 – 1993): PABX and microwave links for a banking network.

EDUCATION

- **42 São Paulo — Software Engineering Cadet** (2025 – present): intensive, project-based, peer-to-peer curriculum in C, algorithms and Unix systems programming
- **Licentiate in Mathematics** (Pedagogical Formation Course in Mathematics) — Universidade Pitágoras Unopar; completed Dec 2020, degree conferred Mar 2021
- **B.S., Electrical Engineering** — Faculdade de Engenharia de São Paulo (FESP), 2002
- **Electronics Technician** — Escola Técnica de Eletrônica Francisco Moreira da Costa, Santa Rita do Sapucaí, MG, 1990

CERTIFICATIONS & TRAINING

- **AWS Certified Cloud Practitioner** — Amazon Web Services (valid through Jun 2028) — credly.com/badges/564e8c2f-bc11-46a9-b1fe-7b0aa87dc545/public_url
- AWS Certified Developer – Associate: in preparation
- AWS re/Start Graduate (May 2025); AWS Educate: Introduction to Generative AI (Mar 2025); Escola da Nuvem (EDN): AWS Fundamentals + AI
- Google Cloud Skill Badges: Prompt Design in Vertex AI (Oct 2024); Inspect Rich Documents with Gemini Multimodality and Multimodal RAG (Nov 2024); Develop GenAI Apps with Gemini and Streamlit (Nov 2024)
- ITIL Service Management — CompanyWeb (2007, 2008); eTOM Distilled — TM Forum
- Ericsson (2006): WCDMA System Overview + CPP Node Features, GPRS System Survey, ATM & IP over ATM, Datacom and IP Fundamentals

LANGUAGES

Portuguese (native) · English (fluent) · Spanish (basic)